

Amal Chakhar, PhD

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Profile

Earth-observation scientist with a PhD and 8+ peer-reviewed publications applying machine learning to satellite imagery for agriculture. Builds end-to-end pipelines on Sentinel-1/-2 and Landsat data — from ingestion and feature engineering to classification and validation — to answer real questions about crops, water and drought.

Skills

Programming: Python (pandas, NumPy, scikit-learn), R, SQL.

Earth Observation: Sentinel-1 (SAR), Sentinel-2, Landsat-8, UAV photogrammetry, time-series analysis, datacubes.

Machine Learning: Supervised classification (Random Forest, SVM, kNN), accuracy assessment, feature engineering on multi-source EO data.

Geospatial: Google Earth Engine, QGIS, ArcGIS, GDAL.

Domain: Crop classification, soil moisture retrieval, irrigation detection, evapotranspiration, drought monitoring.

Languages: Spanish, English, French (professional working proficiency); Arabic (native).

Experience

Postdoctoral Research Fellow, University of Leeds – Leeds, UK

Aug 2023 – Present

- Build Earth observation pipelines for the iSPARK programme — crop classification and drought / irrigation monitoring across western Kenya to support food-security policy decisions.

Doctoral Researcher, University of Castilla-La Mancha – Albacete, Spain

2018 – 2023

- Authored 8 peer-reviewed papers on Sentinel-1/-2 fusion for crop classification, soil moisture retrieval, and irrigation detection (full list at amalchakhar.com).
- Designed a local nested grid combining Landsat-8 and Sentinel-2 — published in *Remote Sensing* (2021).
- Developed open software tooling for spatio-temporal datacubes, applied to crop classification in Cap Bon, Tunisia.

Selected Publications

- [1] **Amal Chakhar** et al. "Irrigation Detection Using Sentinel-1 and Sentinel-2 Time Series on Fruit Tree Orchards". In: *Remote Sensing* 16.3 (2024), p. 458. DOI: 10.3390/rs16030458.
- [2] **Amal Chakhar** et al. "Optimized Software Tools to Generate Large Spatio-Temporal Data Using the Datacubes Concept: Application to Crop Classification in Cap Bon, Tunisia". In: *Remote Sensing* 14.19 (2022), p. 5013. DOI: 10.3390/rs14195013.
- [3] **Amal Chakhar** et al. "Applications of Remote Sensing in Precision Agriculture: A Review". In: *Remote Sensing* 12.19 (2020), p. 3136. DOI: 10.3390/rs12193136.

Education

PhD, Agricultural and Environmental Sciences, University of Castilla-La Mancha, Spain

2023

MSc, Integrated Planning for Rural Development and Environmental Management, Spain

Engineering Degree,

Horticultural Systems, University of Sousse, Tunisia

2015